## Highlights

### How much retrieval gain reaches the answer? Transfer ratios and cost of open embedding models in retrieval-augmented question answering

- On NQ, at most 0.27 EM points per Hit@5 point reached the answer
- On multi-hop HotpotQA, transfer was larger and its intervals reached 1
- Higher EM given a hit carries 48%–70% of the HotpotQA transfer
- Encoding throughput on a free T4 GPU differed 27-fold across six encoders
- Three of six registered predictions failed in at least one condition

# How much retrieval gain reaches the answer? Transfer ratios and cost of open embedding models in retrieval-augmented question answering

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ABSTRACT

Teams that build retrieval-augmented generation (RAG) systems often pick an embedding model by its retrieval score. We measured how much of a retrieval gain reaches the answer, and at what cost, for open models on free cloud GPUs. We compared BM25 and six open embedding models (23–596M parameters) on Natural Questions (NQ, 2,255 queries) and HotpotQA (2,000 queries), each read by two quantised 7–8B models. The transfer ratio $T$ divides the exact-match (EM) gain by the Hit@5 gain over a small baseline encoder on the same queries. We report bootstrap intervals and registered six predictions before the second phase of the study. On NQ, $T$ ranged from 0.16 to 0.27, and no upper 95% bound exceeded 0.45. On HotpotQA, $T$ was larger (0.49–0.93) and one upper bound reached 1.10, so the prediction $T < 1$ failed in one of four reader–dataset cells. An exact decomposition attributes 48%–70% of the HotpotQA transfer to higher EM given a hit, in line with retrieving both supporting passages. On HotpotQA, bge-large gave higher EM than e5-base-v2 ($+0.028$ and $+0.022$, Holm-adjusted $p \leq 0.040$); on NQ, no difference was detected. Encoding throughput on one T4 GPU differed 27-fold across encoders. Top-ranked non-gold passages lowered EM by 0.074 against random passages; gold position showed no detectable effect. An LLM faithfulness judge agreed poorly with human labels (Cohen's $\kappa = 0.127$) and was excluded. Within this setup, retrieval gains overstate answer gains on single-hop questions, and encoder cost should be weighed alongside retrieval scores.

## 1. Introduction

Retrieval-augmented generation (RAG) answers a question in two steps. A retriever selects passages from a corpus, and a language model, the reader, writes an answer from them (Lewis, Perez, Piktus, Petroni, Karpukhin, Goyal, Küttler, Lewis, tau Yih, Rocktäschel, Riedel and Kiela, 2020; Izacard and Grave, 2021; Ram, Levine, Dalmedigos, Muhlgay, Shashua, Leyton-Brown and Shoham, 2023). Teams that build such systems must choose the retriever. For dense retrieval this means choosing a text embedding model, and the usual guide is a retrieval benchmark such as BEIR (Thakur, Reimers, Rücklé, Srivastava and Gurevych, 2021) or MTEB (Muennighoff, Tazi, Magne and Reimers, 2023). These benchmarks score ranked lists against relevance labels. They do not show how much of a retrieval gain survives the reader.

The question matters most for teams with little compute. A larger encoder costs more to run over a corpus and adds latency to every query. If a retrieval gain turns into a much smaller answer gain, a cheaper encoder may serve as well. Several studies report that retrieval improvements do not reliably improve answers (Salemi and Zamani, 2024; Mathur, Rittner, Thakur, Schiff and Islam, 2026; Medrano, Verma and Chhabra, 2026; Ghosh, Mukhopadhyay and Chatterjee, 2026). Others find that retrieval metrics correlate well with generation quality when the two are aligned (Samuel, Martin, Yang, Yates, Lawrie, Soboroff, Dietz and Durme, 2026). Readers are also sensitive to what else is in the context: passages that look relevant but lack the answer can lower accuracy (Cuconasu, Trappolini, Siciliano, Filice, Campagnano, Maarek, Tonellotto and Silvestri, 2024; Amiraz, Cuconasu, Filice and Karnin, 2025; Shi, Chen, Misra, Scales, Dohan, Chi, Schärli and Zhou, 2023).

Earlier studies relate retrieval quality to answer quality through correlations across systems or topics (Salemi and Zamani, 2024; Samuel et al., 2026; Trappolini, Cuconasu, Filice, Maarek and Silvestri, 2025), through a ratio of answer F1 to recall across RAG methods (Peng, Zheng, Hao, Tang, Li, Huang, Huang, Liu, Zhu and E, 2026), or through manipulations of a single retriever (Leto, Aguerrebere, Bhati, Willke, Tepper and Vo, 2024). They do not report, for a set of interchangeable open encoders, the paired ratio of

answer gain to retrieval gain with its uncertainty, per reader and dataset, next to the measured cost of each encoder. This study fills that gap for one well-defined setting.

We define a transfer ratio $T(R)$: the gain in exact match (EM) of retriever $R$ over a small baseline encoder, divided by its gain in Hit@5, both measured on the same queries. A ratio of 1 means every extra question with a relevant passage in the top five also gains a correct answer. A ratio near 0 means the retrieval gain does not reach the answer. We estimate $T(R)$ for BM25 and five dense encoders against MiniLM-L6, with two quantised 7–8B readers, on a single-hop dataset (Natural Questions) and a multi-hop dataset (HotpotQA). All runs used free Kaggle T4 GPUs.

We ask four research questions.

**RQ1** How much of a Hit@5 gain over a small encoder reaches EM, and does this depend on the reader and the dataset?

**RQ2** Do larger or newer encoders give higher EM than a mid-size encoder (e5-base-v2)?

**RQ3** How do context choices change the answer: the number of passages, the kind of non-gold passages, the position of the gold passage, and reranking?

**RQ4** What does each encoder cost on free hardware, and do our sub-corpus and exact search distort the comparison?

The study ran in two phases. Phase 1 covered one reader on NQ. Before Phase 2, we registered six predictions (P1–P6) in a dated design document and then ran the new cells and experiments. We report each prediction as it came out, including the three that failed in at least one condition.

The contributions are as follows.

- A paired estimate of retrieval-to-answer transfer for six retrievers, two readers and two datasets, with bootstrap intervals and a reporting rule fixed in advance (Section 5.2).
- An exact decomposition of the ratio into the hit–miss gap of the baseline and the change in EM given a hit or a miss. It separates the NQ pattern (low transfer) from the HotpotQA pattern (larger transfer driven by the second supporting passage) (Section 5.3).
- Controlled tests of context composition and position, a reranker, top-$k$, approximate search and a full-corpus check, all run in the same pipeline (Sections 5.5–5.6).
- Measured encoding cost of each encoder on a free GPU, with five repetitions, and a public record of every deviation and failed run (Sections 5.6 and 4.1).

Section 2 reviews related work. Sections 3 and 4 describe the method and setup. Section 5 reports the results, Section 6 discusses them, and Section 7 lists the limits of the study.

## 2. Related work

### 2.1. Retrieval quality and answer quality

Dense passage retrieval replaced sparse retrieval as the default initial stage for open-domain question answering (Karpukhin, Oguz, Min, Lewis, Wu, Edunov, Chen and Yih, 2020; Lee, Chang and Toutanova, 2019), and retrieval-augmented readers turned retrieved passages into answers (Lewis et al., 2020; Izacard and Grave, 2021). Sentence encoders and general-purpose embedding models made dense retrieval cheap to adopt (Reimers and Gurevych, 2019; Wang, Yang, Huang, Jiao, Yang, Jiang, Majumder and Wei, 2022; Xiao, Liu, Zhang, Muennighoff, Lian and Nie, 2024; Zhang, Li, Long, Zhang, Lin, Yang, Xie, Yang, Liu, Lin, Huang and Zhou, 2025). Their quality is usually judged on retrieval benchmarks (Thakur et al., 2021; Muennighoff et al., 2023).

How retrieval scores relate to answer scores is studied in three ways. One correlates retrieval metrics with downstream metrics. Salemi and Zamani (2024) report that relevance-label metrics correlate weakly with downstream RAG performance and propose labels derived from the reader itself. Trappolini et al. (2025) argue that rank-discounted metrics fit a human reader, not a language model, and propose a utility-

and distraction-aware gain. Samuel et al. (2026) find positive topic- and system-level correlations between coverage-based retrieval metrics and the coverage of generated reports, with weaker links for relevance-based nDCG. A second builds ratios. Peng et al. (2026) define a recall conversion rate as answer F1 divided by Recall@5 and report that it falls as recall rises across RAG methods. Their ratio compares levels across different pipelines. Our ratio compares paired gains of interchangeable encoders against one baseline. A third manipulates retrieval directly. Leto et al. (2024) vary the share of gold passages and the accuracy of approximate search for one dense retriever and find that gold-document recall matters far more than search recall. Domain studies report retrieval gains that did not improve answers (Mathur et al., 2026; Medrano et al., 2026; Muthyam, 2026; Ghosh et al., 2026). Bala (2026) show that recall of the retrieved set can differ from what reaches the reader once a context budget forces truncation.

### 2.2. Context composition

Readers react to the passages next to the gold one. Shi et al. (2023) show that irrelevant context distracts language models. Cuconasu et al. (2024) find that high-scoring passages without the answer lower accuracy, while random passages placed before the gold passage can raise it. Amiraz et al. (2025) quantify a distracting effect per passage. Jin, Yoon, Han and Arik (2024) attribute the drop in quality at large retrieval depth to hard negatives, and Repantis, Gawde, Singh, Alekar, Zhang, Karslioglu and Vishwakarma (2026) isolate competition between passages from context length. Position effects were reported for long contexts (Liu, Lin, Hewitt, Paranjape, Bevilacqua, Petroni and Liang, 2024), but Cuconasu, Filice, Horowitz, Maarek and Silvestri (2025) find them marginal in realistic retrieval settings. Hsia, Shaikh, Wang and Neubig (2025) report that the reader's tolerance of noise shapes how a RAG system scales with retrieval depth, and Zhao, Shao, Huang, Song, Wang, Wan and Ma (2026) find that 5–10 documents work best for question answering across several models.

### 2.3. Encoder cost

Embedding benchmarks report quality; cost is rarely measured in the same study. Assadi, Muennighoff and Lee (2026) compare the cost of embedding models and language models on embedding tasks, without a reader. Approximate nearest-neighbour search reduces query cost (Malkov and Yashunin, 2020; Johnson, Douze and Jegou, 2021); Leto et al. (2024) find that it changes answer quality little. Efficient serving (Kwon, Li, Zhuang, Sheng, Zheng, Yu, Gonzalez, Zhang and Stoica, 2023) and weight quantisation (Lin, Tang, Tang, Yang, Chen, Wang, Xiao, Dang, Gan and Han, 2024) make 7–8B readers fit on one consumer-grade GPU.

### 2.4. Evaluation practice

RAG evaluation frameworks use language models as judges (Es, James, Espinosa Anke and Schockaert, 2024; Saad-Falcon, Khattab, Potts and Zaharia, 2024; Zheng, Chiang, Sheng, Zhuang, Wu, Zhuang, Lin, Li, Li, Xing, Zhang, Gonzalez et al., 2023), and benchmarks package many retrievers and readers (Rau, Déjean, Chirkova, Formal, Wang, Clinchant and Nikoulina, 2024; Petroni, Piktus, Fan, Lewis, Yazdani, De Cao, Thorne, Jernite, Karpukhin, Maillard, Plachouras, Rocktäschel et al., 2021; Lyu, Li, Niu, Xiong, Tang, Wang, Wu, Liu, Xu and Chen, 2025). Lexical EM undercounts correct answers from language models (Kamalloo, Dziri, Clarke and Rafiei, 2023); we therefore also report token F1 and answer containment. For significance testing we follow advice from natural language processing and information retrieval: paired tests, multiplicity control and effect sizes with intervals (Dror, Baumer, Shlomov and Reichart, 2018; Smucker, Allan and Carterette, 2007; Sakai, 2014).

## 3. Method

### 3.1. Transfer ratio

Let $Q$ be a fixed set of generation queries. For retriever $R$ and query $q$, let $h_R(q) \in \{0,1\}$ indicate that at least one gold passage of $q$ is among the top five passages, and let $e_R(q) \in \{0,1\}$ be the exact match of the reader's answer given those five passages. Write $\bar{h}_R$ and $\bar{e}_R$ for the means over $Q$. With MiniLM-L6 as the baseline $R_1$, the transfer ratio is

$$T(R) = \frac{\bar{e}_R - \bar{e}_{R_1}}{\bar{h}_R - \bar{h}_{R_1}}. \tag{1}$$

Both differences are paired: they use the same queries, the same reader, the same prompt and the same decoding. $T(R)$ is a ratio of mean differences. It is not the probability that one extra hit produces one extra correct answer, because the queries that gain a hit need not be the queries that gain an answer. We report $T(R)$ only if $|\bar{h}_R - \bar{h}_{R_1}|$ exceeds 2 percentage points and the Hit@5 difference is significant after Holm correction; otherwise the ratio is marked as not estimable. This rule was fixed in the Phase 1 design to avoid ratios with near-zero denominators.

*Decomposition* Let $a_R = P(e_R = 1 \mid h_R = 1)$ and $b_R = P(e_R = 1 \mid h_R = 0)$ be the EM rates among hits and misses. Then $\bar{e}_R = \bar{h}_R a_R + (1 - \bar{h}_R) b_R$, and Eq. 1 splits exactly into

$$T(R) = \underbrace{(a_{R_1} - b_{R_1})}_{\text{baseline hit–miss gap}} + \underbrace{\frac{\bar{h}_R\,(a_R - a_{R_1}) + (1 - \bar{h}_R)\,(b_R - b_{R_1})}{\bar{h}_R - \bar{h}_{R_1}}}_{\text{change in conditional EM}}. \tag{2}$$

If a better encoder only turned misses into hits and left $a$ and $b$ unchanged, $T(R)$ would equal the baseline gap. The second term shows how far the encoder also changes how useful its hits and misses are. This decomposition was not part of the registered analysis; we report it as a post hoc description.

### 3.2. Data

*Natural Questions* We used the BEIR version of Natural Questions (Kwiatkowski, Palomaki, Redfield, Collins, Parikh, Alberti, Epstein, Polosukhin, Devlin, Lee, Toutanova, Jones et al., 2019; Thakur et al., 2021): 3,452 judged test queries and a corpus of 2,681,468 Wikipedia passages. BEIR does not ship short answers, so we joined its queries to NQ-open (Lee et al., 2019) by normalised question text. The join matched 2,255 queries; 1,197 had no match and none had more than one. All 2,255 joined queries form the NQ generation set.

*HotpotQA* We used the BEIR version of HotpotQA (Yang, Qi, Zhang, Bengio, Cohen, Salakhutdinov and Manning, 2018; Thakur et al., 2021): 7,405 judged test queries, two gold passages per query, and a corpus of 5,233,329 passages. Answers come from the fullwiki validation release, joined by question id; all queries matched. The generation set is the leading 2,000 queries of a seeded order (1616 bridge and 384 comparison questions).

*Sub-corpus* Encoding millions of passages with six encoders exceeds a free GPU budget, so each dataset uses a 300,000-passage sub-corpus. It contains the gold passages of all judged queries, BM25 top-$d$ non-gold passages for every judged query ($d = 42$ for NQ and 19 for HotpotQA, set by a pre-registered formula), and a seeded random fill. The NQ sub-corpus has 4,201 gold, 127,791 BM25 and 168,008 random passages; the HotpotQA sub-corpus has 13,783, 118,726 and 167,491. Because every gold passage is present, absolute retrieval scores are higher than on the full corpus. Section 5.6 checks whether the ordering of retrievers changes.

### 3.3. Retrievers

BM25 used the bm25s library (Lù, 2024) with the Lucene variant, $k_1 = 1.2$, $b = 0.75$, lower-casing and no stemming or stop-word removal (Robertson and Zaragoza, 2009). The six dense encoders were all-MiniLM-L6-v2 (Reimers and Gurevych, 2019; Wang, Wei, Dong, Bao, Yang and Zhou, 2020), bge-small, bge-base and bge-large v1.5 (Xiao et al., 2024), e5-base-v2 (Wang et al., 2022) and Qwen3-Embedding-0.6B (Zhang et al., 2025). Each model ran at a pinned revision in half precision with a maximum of 256 tokens and its documented query and passage prefixes. Dense search used exact inner-product search in FAISS (Johnson et al., 2021). Table 7 lists parameter counts.

### 3.4. Readers and prompts

The readers were Qwen2.5-7B-Instruct (Qwen, Yang, Yang, Zhang, Hui, Zheng, Yu, Li, Liu, Huang, Wei, Lin et al., 2024) and Llama-3.1-8B-Instruct (Grattafiori, Dubey, Jauhri, Pandey, Kadian, Al-Dahle, Letman, Mathur, Schelten, Vaughan, Yang, Fan et al., 2024), both in 4-bit AWQ form (Lin et al., 2024), served with vLLM (Kwon et al., 2023) on one T4 each. Decoding was greedy with at most 32 new tokens and a stop at

a newline. The main prompt (A1) gives the top $k = 5$ passages in rank order, each as "[i] title / text", and asks for a short answer (Appendix A). Two reference conditions bound each cell: closed book (no passages) and gold passages only.

### 3.5. Metrics

Hit@5 is 1 if at least one gold passage is in the top five. For HotpotQA we also report Both@5, which requires both gold passages. nDCG@10 uses binary gains. EM is 1 if the normalised answer equals a normalised reference answer. Normalisation follows SQuAD (lower case, no punctuation or articles, single spaces) and also removes typographic quotes, en and em dashes and the ellipsis; a plain SQuAD normaliser would change the score of 91 of 131,855 answers (0.07%). Token F1 and answer containment (AC; a normalised reference occurs in the normalised output) are secondary metrics, because lexical EM undercounts correct answers from language models (Kamalloo et al., 2023).

### 3.6. Statistical analysis

All comparisons are paired by query. Binary outcomes use the exact McNemar test (McNemar, 1947). Within each reader–dataset cell, the 21 pairwise retriever comparisons of a metric form one family with Holm correction (Holm, 1979). Intervals are 95% percentile bootstrap intervals over queries (Efron and Tibshirani, 1993) with 10,000 resamples and seed 0; for $T(R)$ the numerator and denominator are resampled together. A secondary model pools the four cells: a logistic generalised estimating equation (Liang and Zeger, 1986) for EM with retriever, dataset and reader as factors, an exchangeable working correlation within query, and empirical (sandwich) standard errors. Agreement between the LLM judge and human labels uses Cohen's $\kappa$ (Cohen, 1960; Landis and Koch, 1977). A non-significant test is reported as "no difference detected", not as equivalence.

### 3.7. Pre-registered predictions

Phase 1 (Qwen reader, NQ) was designed and committed before any data were generated. Before Phase 2 we registered six predictions (Table 11):

**P1** In each new cell, every reportable $T(R)$ of a dense encoder has an upper 95% bound below 1.

**P2** In each new cell, bge-large and Qwen3-Embedding-0.6B show no Holm-significant EM gain over e5-base-v2.

**P3** The nDCG@10 ordering of BM25, MiniLM-L6, bge-small and e5-base-v2 on the sub-corpus holds on the full corpus.

**P4** For every retriever, EM at $k = 1$ is lower than at $k = 5$.

**P5** Gold plus four hard negatives gives lower EM than gold plus four random passages.

**P6** The position of the gold passage (1, 3 or 5) changes EM.

The Qwen–NQ cell and the $k$ ablation for BM25, MiniLM-L6 and bge-large existed before registration. Their P1, P2 and P4 verdicts are therefore not confirmatory, and Table 11 marks them. All other analyses are descriptive or secondary.

### 3.8. Additional experiments

*Top-k (E4)* The Qwen reader answered all NQ queries with $k \in \{1, 3, 5, 10\}$ for all seven retrievers.

*Context composition (E12)* On the leading 500 NQ generation queries of the seeded order, the Qwen reader always saw five passages: gold at position 1 plus four hard negatives (M1); gold at position 3 or 5 among the same negatives (M3, M5); gold at position 1 plus four random passages (MR); and the gold passage alone (MG). Hard negatives are the top-ranked BM25 passages that are not labelled gold.

*Reranking (E5)* A cross-encoder, bge-reranker-base, reranked the top 50 passages of BM25 and of e5-base-v2 (Nogueira and Cho, 2019; Xiao et al., 2024); the Qwen reader then answered NQ from the reranked top five.

*Approximate search (E6)* HNSW indexes (Malkov and Yashunin, 2020) ($M = 32$, efConstruction 200, efSearch 64 and 128) replaced exact search for MiniLM-L6, bge-small and e5-base-v2 on the NQ sub-corpus.

*Full corpus (E9)* BM25, MiniLM-L6, bge-small and e5-base-v2 retrieved from the full NQ corpus for all judged queries.

*Cost (E7)* For each encoder we measured passage-encoding throughput on 20,000 passages (batch 128, after warm-up, with GPU synchronisation) and batch-1 query-encoding latency on 500 queries, five times in one session.

*Judge validation (E3)* In Phase 1, an LLM judge (Mistral-7B-Instruct-v0.3; Jiang, Sablayrolles, Mensch, Bamford, Chaplot, de las Casas, Bressand, Lengyel, Lample, Saulnier, Lavaud, Lachaux et al., 2023) labelled whether each answer was supported by its passages, for 500 NQ queries per retriever. The author labelled a stratified sample of 200 items without seeing the judge's verdicts. The registered rule was: use the judge if $\kappa \geq 0.4$, use it only with a sensitivity analysis if $0.2 \leq \kappa < 0.4$, and exclude it if $\kappa < 0.2$.

## 4. Experimental setup

*Hardware and software* All runs used free Kaggle notebook sessions with two NVIDIA Tesla T4 GPUs (15 GB each), four virtual CPUs and 31 GB of memory. Each process used one GPU; two processes could run in parallel. The code of every session was embedded in its notebook and checked against a fixed version-control tree hash before it ran. Every session recorded its Python version, package versions and model revisions. The readers ran with the same stack in every session (PyTorch 2.13.0 with CUDA 13.0, vLLM 0.30.0, Transformers 5.17.0). Between the two phases the Kaggle image changed from Python 3.12.13 to 3.13.15, and NumPy from 2.0.2 to 2.1.3.

*Determinism across phases* Some comparisons cross the two phases: the Qwen–NQ cell against the new cells in the pooled model, $k = 5$ against the new $k$ values, and the reranker runs against the Phase 1 runs. To check that the software change does not alter the reader, we compared the two phases on inputs that were identical by construction: the gold-only context of Phase 1 and the gold-only condition of the context experiment. Of 397 identical prompts, 396 produced identical outputs; the one difference was a shorter answer to the same question. A rerun of 100 prompts within one session gave 100/100 identical outputs.

*Seeds and repetitions* The query order (seed 20260926), bootstrap (seed 0), decoding (greedy, seed 0) and the random passages of the context experiment were fixed. Retrieval and greedy decoding are deterministic within a session, so we ran each answer condition once. The cost measurements were repeated five times.

*Analysis and verification* One analysis script reads only the committed raw outputs and writes all statistics, tables and figures. The author reran it on a second computer and obtained identical output. A separate script, written without the project code and using only the Python standard library, re-scored all 131,855 answers from the raw reader outputs, recomputed every Hit@5 and EM mean, every transfer ratio point estimate, all 84 pairwise EM tests with their Holm adjustment, and Cohen's $\kappa$. It found no disagreement. Bootstrap intervals, F1, answer containment and the pooled model were not re-implemented independently.

### 4.1. Deviations and failed runs

We list every departure from the registered design.

1. **NQ join gate.** The original design required a join rate of at least 0.70 between BEIR-NQ and NQ-open; the observed rate was 0.653. Before any outcome was observed, we replaced the gate with a minimum of 1,000 joined queries, because the two resources are independently filtered subsets of Natural Questions. Retrieval results on all 3,452 judged queries are reported where they matter (Table 9).
2. **Software image.** The initial Phase 2 session stopped at the environment check because the Kaggle image had moved to Python 3.13, for which the pinned NumPy had no wheel. We unpinned NumPy and recorded all version differences; no data were produced by the stopped run.

3. **Failed reader start.** The initial HotpotQA reader session stopped before generating anything because the image's torchaudio build did not match the CUDA version of PyTorch. Later sessions reinstalled the matching build and checked the stack before any other step.
4. **Reduced scope of E6.** Approximate search was planned for all six encoders. To stay within the GPU quota it ran for MiniLM-L6, bge-small and e5-base-v2, using the sub-corpus rows of the full-corpus embeddings.
5. **E7 scope.** The repeated cost measurement covered encoding throughput and query-encoding latency; search latency was measured separately in E6 and is not repeated.
6. **Failure log.** The design promised one machine-readable failure log. Failures were recorded instead in two dated design addenda and in the session logs.

No run was repeated to obtain a different result, and no result was discarded.

## 5. Results

### 5.1. Retrieval

Table 1 gives Hit@5 on the generation queries; Table 13 gives the intervals. On NQ, the dense encoders spread widely. MiniLM-L6 reached 0.664 and e5-base-v2 0.816; bge-large (0.796) came second. Qwen3-Embedding-0.6B (0.744) placed between bge-small and bge-base. BM25 reached 0.363. On HotpotQA, Hit@5 nearly saturated: all dense encoders except MiniLM-L6 scored between 0.945 and 0.961, against 0.817 for MiniLM-L6 and 0.833 for BM25. The stricter Both@5 separated them more clearly: bge-large found both supporting passages for 0.621 of the questions, e5-base-v2 for 0.544, Qwen3-Embedding-0.6B for 0.444 and MiniLM-L6 for 0.230.

### 5.2. Answers and transfer (RQ1)

Table 1 gives EM for every cell with its two bounds, and Fig. 1 plots EM against Hit@5. Retrieval closed only part of the gap between the closed-book and gold-passage conditions. With Qwen on NQ, closed-book EM was 0.149, gold passages gave 0.653, and the best retriever, e5-base-v2, gave 0.510. With Llama on NQ the best retriever was bge-base (0.477) against a gold bound of 0.613. On HotpotQA, bge-large gave the highest EM for both readers (0.424 and 0.430), against gold bounds of 0.628 and 0.624.

Table 2 and Fig. 2 give the transfer ratios. On NQ, the dense encoders had ratios between 0.16 (bge-large and e5-base-v2 with Llama) and 0.27 (bge-base with Llama). With Qwen, e5-base-v2 gained +0.153 in Hit@5 over MiniLM-L6 but only +0.038 in EM, a ratio of 0.25 [0.15, 0.36]. The intervals for bge-small included zero in both NQ cells (Qwen [−0.07, 0.44], Llama [−0.08, 0.43]), so its NQ ratio is uncertain in sign. No upper bound on NQ exceeded 0.45. BM25 retrieved worse than MiniLM-L6 on NQ, so its ratio (0.37 with Qwen) describes how much of the retrieval loss reached the answer.

On HotpotQA the ratios were larger for every encoder and reader. They ranged from 0.49 (Qwen3-Embedding-0.6B with Qwen) to 0.93 (bge-large with Llama). With Qwen, every upper bound stayed below 1, the largest being 0.996 for bge-large. With Llama, the bounds were 1.000 for bge-base, 1.000 for e5-base-v2 and 1.096 for bge-large; three of five bounds reached or exceeded 1. Prediction P1 therefore held in the two new NQ and Qwen–HotpotQA cells and failed in the Llama–HotpotQA cell (Table 11). The BM25 ratio was not estimable on HotpotQA, because BM25 and MiniLM-L6 differed by only +0.016 in Hit@5.

The lenient answer metrics moved the ratios up but did not change the pattern (Table 14). With token F1, the NQ ratios of the dense encoders stayed between 0.23 and 0.33. On HotpotQA with Llama, the F1 ratio of bge-large was 1.10 [0.95, 1.28], so under F1 a full transfer cannot be ruled out for that encoder.

### 5.3. Where the transfer comes from

The decomposition in Eq. 2 separates two effects (Table 15). One term, the hit–miss gap of MiniLM-L6, was similar in all cells: 0.33 and 0.32 on NQ, 0.25 and 0.28 on HotpotQA. The other term differed in sign between the datasets.

On NQ it was negative for every dense encoder (from −0.16 to −0.05 with Llama; from −0.14 to −0.08 with Qwen). Better encoders found gold passages for more questions, but EM given a hit fell slightly (Qwen: 0.584 for MiniLM-L6, 0.569 for e5-base-v2). EM given a miss stayed near 0.244, well above closed-book EM (0.149). A "miss" under the sparse NQ labels thus often still produced a correct answer.

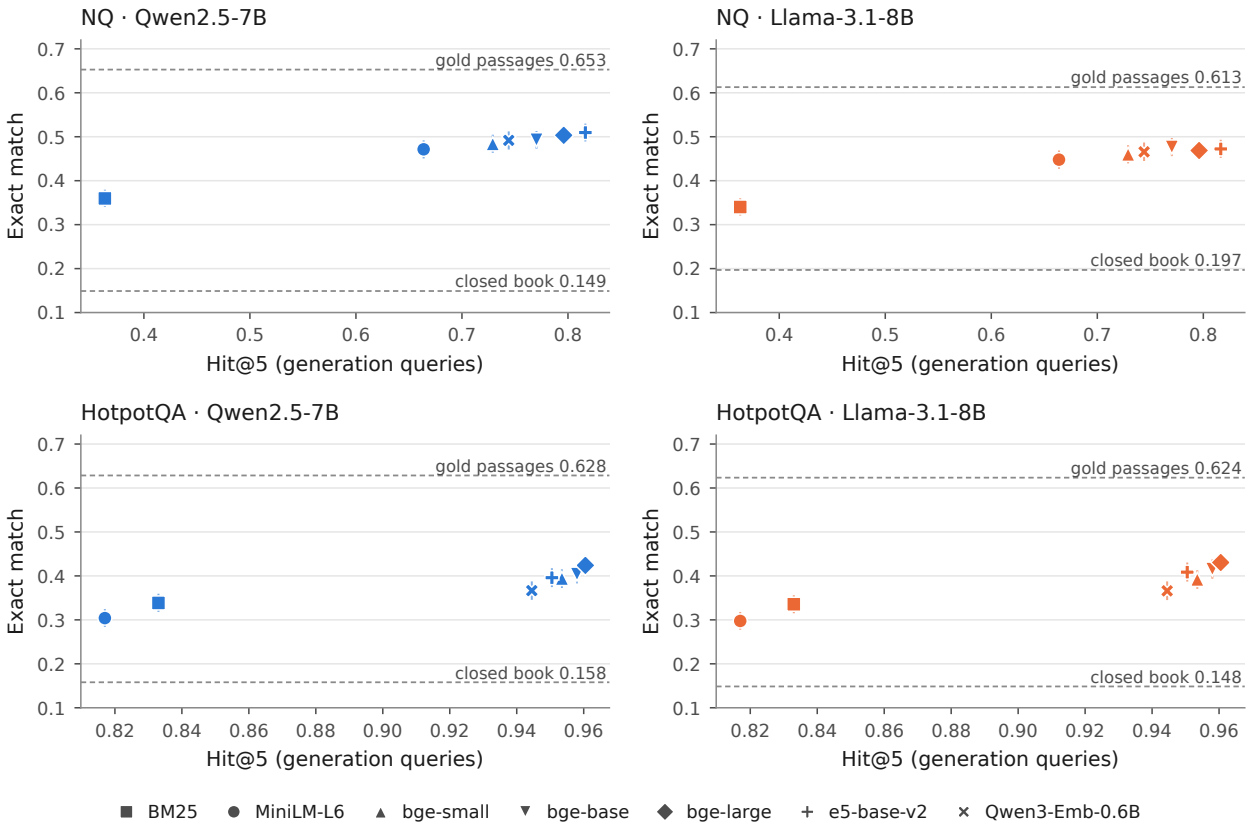



**Figure 1**: Exact match against Hit@5 for each retriever in the four reader–dataset cells. Vertical bars are 95% bootstrap intervals for EM. Dashed lines mark the closed-book and gold-passage conditions. Note the different Hit@5 ranges of the two datasets.

On HotpotQA the second term was positive and large, from +0.24 (Qwen3-Embedding-0.6B with Qwen) to +0.65 (bge-large with Llama). It made up 48% to 70% of the ratio. EM given a hit rose with the encoder (Qwen: 0.350 for MiniLM-L6, 0.438 for bge-large), while EM given a miss was lower for the stronger encoders (Qwen: 0.098 for MiniLM-L6, 0.030 to 0.090 for the others). The rise matches the Both@5 results: better encoders more often retrieved the second supporting passage, which Hit@5 does not count. In a post hoc check, the EM ranking of the seven retrievers equalled their Both@5 ranking in both HotpotQA cells (Spearman $\rho = 1.00$ and 1.00); against Hit@5 it was 0.96 and 0.96. Bridge questions drove this. With Qwen, bridge EM rose from 0.248 (MiniLM-L6) to 0.384 (bge-large), while comparison EM moved from 0.539 to 0.594, close to its gold bound of 0.654.

## 5.4. Encoder comparisons (RQ2)

Prediction P2 held on NQ and failed on HotpotQA (Table 11). On NQ, neither bge-large nor Qwen3-Embedding-0.6B differed detectably from e5-base-v2 in EM with either reader. On HotpotQA, bge-large gave higher EM than e5-base-v2 with both readers (+0.028, Holm-adjusted $p = 0.0055$; +0.022, $p = 0.040$), in line with its higher Both@5. Qwen3-Embedding-0.6B gave lower EM than e5-base-v2 on HotpotQA (−0.030, $p = 0.0031$; −0.042, $p = 8.5 \times 10^{-6}$). The pooled model over all four cells (Table 3; 59,570 answers in 4,255 query clusters) gave the same picture: odds ratio 1.038 [1.001, 1.076] for bge-large and 0.909 [0.875, 0.943] for Qwen3-Embedding-0.6B, each against e5-base-v2. The bge-large advantage was small and its interval nearly touched 1. The pooled model mixes the two study phases and is secondary.

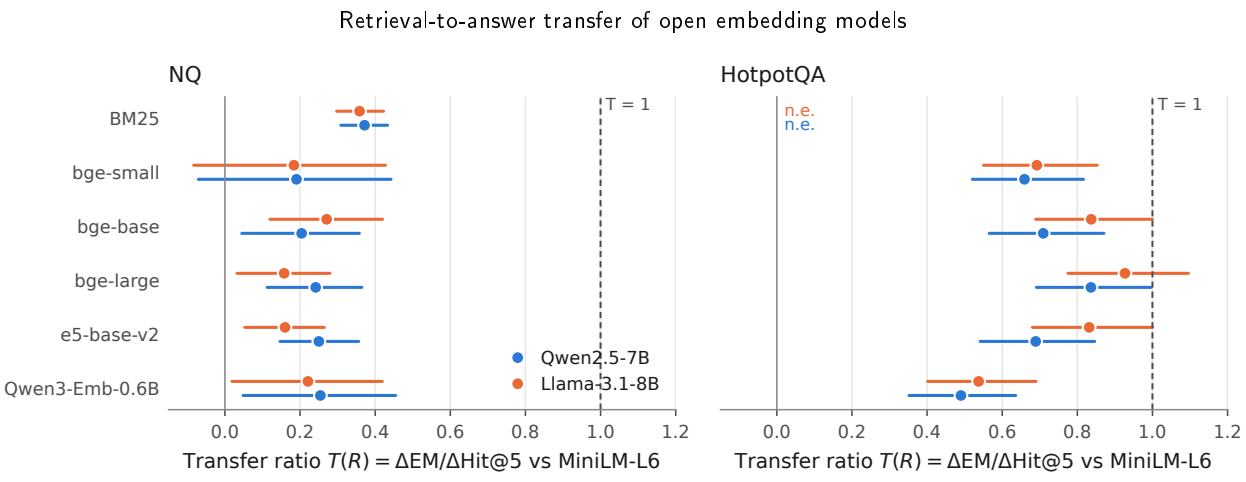



**Figure 2**: Transfer ratio $T(R) = \Delta\mathrm{EM}/\Delta\mathrm{Hit@5}$ against MiniLM-L6 with 95% bootstrap intervals. The dashed line marks $T = 1$. n.e.: not estimable under the reporting rule.

**Table 1**
Retrieval (Hit@5) and answer quality (exact match) for each reader and dataset. Hit@5 is computed on the generation queries; 95% bootstrap intervals are in Table 13.

| | NQ (2,255 queries) | | | | HotpotQA (2,000 queries) | | | |
|---|---|---|---|---|---|---|---|---|
| | Qwen | | Llama | | Qwen | | Llama | |
| Retriever | Hit@5 | EM | Hit@5 | EM | Hit@5 | EM | Hit@5 | EM |
| BM25 | 0.363 | 0.360 | 0.363 | 0.340 | 0.833 | 0.339 | 0.833 | 0.336 |
| MiniLM-L6 | 0.664 | 0.471 | 0.664 | 0.448 | 0.817 | 0.304 | 0.817 | 0.297 |
| bge-small | 0.729 | 0.484 | 0.729 | 0.460 | 0.954 | 0.394 | 0.954 | 0.392 |
| bge-base | 0.770 | 0.493 | 0.770 | 0.477 | 0.958 | 0.404 | 0.958 | 0.415 |
| bge-large | 0.796 | 0.503 | 0.796 | 0.469 | 0.961 | 0.424 | 0.961 | 0.430 |
| e5-base-v2 | 0.816 | 0.510 | 0.816 | 0.472 | 0.951 | 0.396 | 0.951 | 0.408 |
| Qwen3-Emb-0.6B | 0.744 | 0.492 | 0.744 | 0.466 | 0.945 | 0.366 | 0.945 | 0.366 |
| Closed book | – | 0.149 | – | 0.197 | – | 0.158 | – | 0.148 |
| Gold passages | – | 0.653 | – | 0.613 | – | 0.628 | – | 0.624 |

**Table 2**
Transfer ratio $T(R) = \Delta\mathrm{EM}/\Delta\mathrm{Hit@5}$ relative to MiniLM-L6, with 95% bootstrap intervals.

| Retriever | NQ Qwen | NQ Llama | HotpotQA Qwen | HotpotQA Llama |
|---|---|---|---|---|
| BM25 | 0.37 [0.31, 0.43] | 0.36 [0.30, 0.42] | n.e. | n.e. |
| bge-small | 0.19 [−0.07, 0.44] | 0.18 [−0.08, 0.43] | 0.66 [0.52, 0.82] | 0.69 [0.55, 0.85] |
| bge-base | 0.20 [0.04, 0.36] | 0.27 [0.12, 0.42] | 0.71 [0.57, 0.87] | 0.84 [0.69, 1.00] |
| bge-large | 0.24 [0.11, 0.36] | 0.16 [0.03, 0.28] | 0.84 [0.69, 1.00] | 0.93 [0.78, 1.10] |
| e5-base-v2 | 0.25 [0.15, 0.36] | 0.16 [0.05, 0.26] | 0.69 [0.54, 0.85] | 0.83 [0.68, 1.00] |
| Qwen3-Emb-0.6B | 0.25 [0.05, 0.45] | 0.22 [0.02, 0.42] | 0.49 [0.35, 0.64] | 0.54 [0.40, 0.69] |

n.e.: not estimable under the pre-registered rule ($|\Delta\mathrm{Hit@5}| \leq 2$ points or not significant after Holm). BM25 on NQ retrieves worse than MiniLM-L6, so its ratio describes a loss.

## 5.5. Context choices (RQ3)

*Number of passages* Table 4 and Fig. 3 show EM for $k \in \{1, 3, 5, 10\}$ with Qwen on NQ. One passage gave lower EM than five for every retriever (P4 held; differences from −0.126 for BM25 to −0.078 for e5-base-v2, all Holm-adjusted $p < 0.001$). From five to ten passages, EM rose for six retrievers and stayed flat for

**Table 3**
Pooled GEE logistic model of EM over the four reader–dataset cells.

| Contrast | Odds ratio [95% CI] | $p$ |
|---|---|---|
| bge-large vs e5-base-v2 | 1.038 [1.001, 1.076] | 0.043 |
| Qwen3-Emb-0.6B vs e5-base-v2 | 0.909 [0.875, 0.943] | <0.001 |

EM ∼ retriever + dataset + reader; exchangeable working correlation (0.60) within query; 59,570 answers in 4,255 query clusters; empirical (sandwich) standard errors.

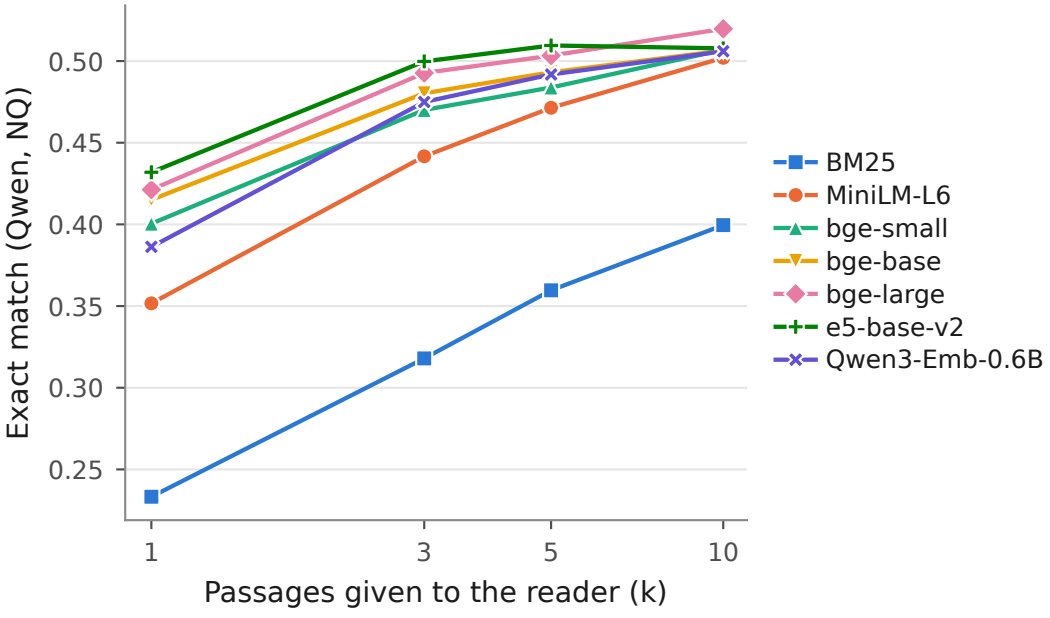



**Figure 3**: Exact match by number of passages $k$, Qwen reader on NQ (2,255 queries).

e5-base-v2 (0.510 and 0.508). The gap between retrievers shrank with $k$: at $k = 10$ all dense encoders gave EM between 0.502 and 0.520.

*Passage composition and position* Table 5 and Fig. 4 give the controlled context test on 500 NQ questions. With the gold passage at position 1, four top-ranked non-gold BM25 passages gave EM 0.580, and four random passages gave 0.654: a drop of 0.074 (exact McNemar $p = 2.2 \times 10^{-5}$; P5 held). Random passages were harmless: gold at position 1 plus four random passages did not differ detectably from the gold passage alone (0.648; $p = 0.71$). The position of the gold passage among the hard negatives showed no detectable effect (EM 0.580, 0.564 and 0.578 at positions 1, 3 and 5; all Holm-adjusted $p \geq 0.94$; P6 failed).

*Reranking* The cross-encoder helped BM25 and hurt e5-base-v2 (Table 6). On the NQ generation queries, reranking the BM25 top 50 raised Hit@5 by +0.304 and EM by +0.097 (both $p < 0.001$). Reranking the e5-base-v2 top 50 lowered Hit@5 by −0.067 ($p = 2.6 \times 10^{-12}$); for EM the change was −0.016 ($p = 0.069$), and no difference was detected. Reranking cost a median of 208 to 249 ms per query on a T4. These results come from one reranker at one depth.

### 5.6. Cost and checks of the setup (RQ4)

*Encoding cost* Table 7 gives the measured cost. Throughput fell from 1,807 passages per second for MiniLM-L6 to 66 for Qwen3-Embedding-0.6B, a factor of 27. e5-base-v2 encoded 2.9 times faster than bge-large. The five repetitions varied little (Table 7): their throughput ranges did not overlap for any pair of encoders except bge-base and e5-base-v2, which share the same architecture size. Query encoding took 6.0 to 52.2 ms

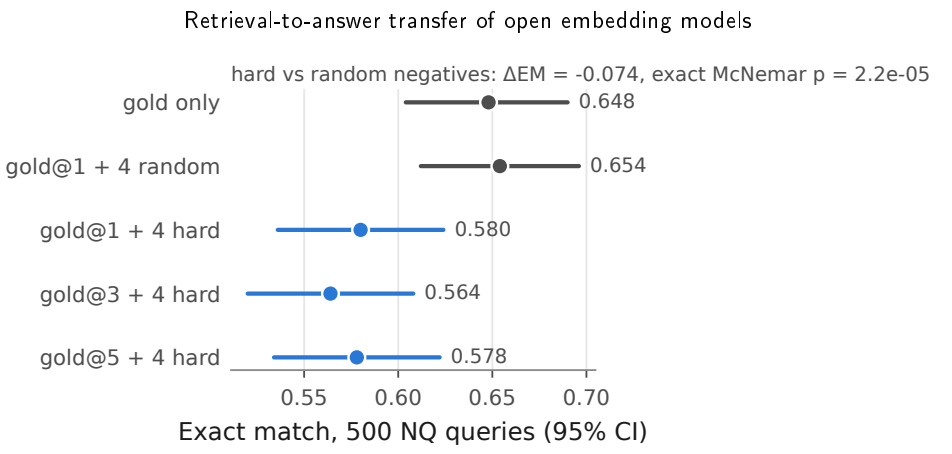



**Figure 4:** Exact match for five controlled contexts of five passages (Qwen reader, 500 NQ generation queries) with 95% bootstrap intervals.

**Table 4**
Exact match by number of passages $k$ (Qwen reader, NQ, 2,255 queries).

| Retriever | $k$=1 | $k$=3 | $k$=5 | $k$=10 | $\Delta_{1-5}$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|---|
| BM25 | 0.233 | 0.318 | 0.360 | 0.400 | −0.126 | <0.001 |
| MiniLM-L6 | 0.352 | 0.442 | 0.471 | 0.502 | −0.120 | <0.001 |
| bge-small | 0.400 | 0.470 | 0.484 | 0.507 | −0.083 | <0.001 |
| bge-base | 0.415 | 0.480 | 0.493 | 0.506 | −0.078 | <0.001 |
| bge-large | 0.421 | 0.493 | 0.503 | 0.520 | −0.082 | <0.001 |
| e5-base-v2 | 0.432 | 0.500 | 0.510 | 0.508 | −0.078 | <0.001 |
| Qwen3-Emb-0.6B | 0.386 | 0.475 | 0.492 | 0.506 | −0.106 | <0.001 |

Exact McNemar test of $k$=1 vs $k$=5, Holm over seven retrievers.

**Table 5**
Distraction and position tests (Qwen reader, 500 NQ generation queries).

| Context (5 passages) | EM [95% CI] | F1 | AC |
|---|---|---|---|
| Gold only | 0.648 [0.604, 0.690] | 0.772 | 0.770 |
| Gold@1 + 4 random | 0.654 [0.612, 0.696] | 0.777 | 0.764 |
| Gold@1 + 4 hard | 0.580 [0.536, 0.624] | 0.696 | 0.686 |
| Gold@3 + 4 hard | 0.564 [0.520, 0.608] | 0.684 | 0.672 |
| Gold@5 + 4 hard | 0.578 [0.534, 0.622] | 0.698 | 0.682 |

Hard vs random negatives: $\Delta$EM = −0.074, exact McNemar $p = 2.2 \times 10^{-5}$. Position: no pair differs after Holm.

per query; BM25 took 25.7 ms on the CPU. Fig. 5 relates throughput to EM averaged over the four cells. bge-large had the highest average EM (0.457), bge-base and e5-base-v2 followed at 0.447 with 2.9 times its throughput, and Qwen3-Embedding-0.6B (0.422) was both the slowest and below bge-small (0.432).

*Approximate search* HNSW search kept the exact top five almost unchanged (Table 8). At efSearch 128 the top-5 overlap with exact search was 0.9956 or higher, Hit@5 changed by at most 0.0023, and the median query latency on the CPU fell from 36–78 ms to 0.84–1.45 ms.

*Sub-corpus versus full corpus* On the full NQ corpus the four tested retrievers kept their nDCG@10 order (Kendall's $\tau = 1.00$; P3 held; Table 9 and Fig. 6). The dense retrievers lost 0.083 to 0.086 in nDCG@10, and BM25 lost almost nothing (0.002). The sub-corpus therefore inflates absolute dense scores but not their order. Its BM25-mined negatives may also make the sub-corpus easier for BM25 relative to the dense encoders than the full corpus is; we did not test reader answers on the full corpus.

**Table 6**
Cross-encoder reranking (bge-reranker-base, top 50) on NQ.

| System | Hit@5 | nDCG@10 | EM | ms/query |
|---|---|---|---|---|
| BM25 | 0.373 | 0.284 | 0.360 | – |
| BM25 + reranker | 0.658 | 0.542 | 0.456 | 208 |
| e5-base-v2 | 0.817 | 0.665 | 0.510 | – |
| e5-base-v2 + reranker | 0.742 | 0.601 | 0.494 | 249 |

Retrieval on all 3,452 judged queries; EM on the 2,255 generation queries (Qwen). Latency: median per query, 50 pairs, one T4.

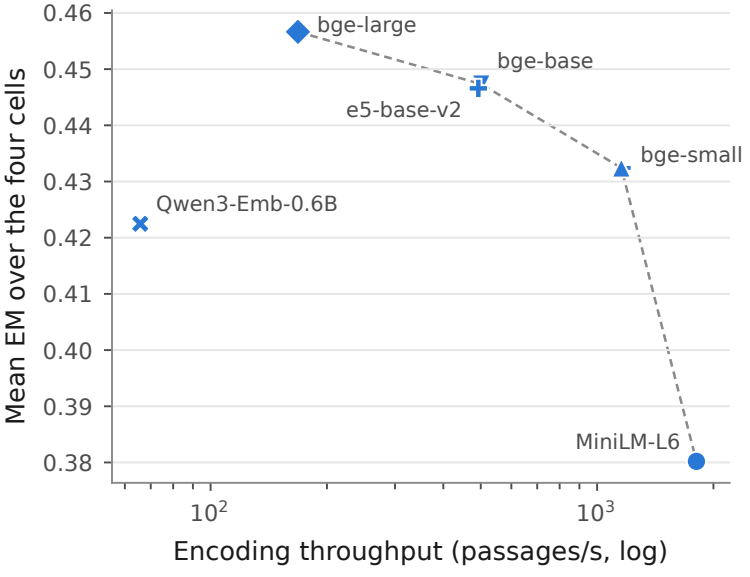



**Figure 5**: Encoding throughput on one T4 GPU (median of five repetitions, log scale) against EM averaged over the four reader–dataset cells. The dashed line joins encoders that no other encoder beats on both axes.

**Table 7**
Encoding cost on one Tesla T4, median (range) over five repetitions.

| Encoder | Params (M) | Dim | Passages/s | Query ms |
|---|---|---|---|---|
| MiniLM-L6 | 23 | 384 | 1807 (1768–1863) | 6.0 (6.0–6.1) |
| bge-small | 33 | 384 | 1155 (1136–1207) | 10.3 (10.1–10.3) |
| bge-base | 109 | 768 | 502 (501–503) | 10.0 (9.9–10.1) |
| e5-base-v2 | 109 | 768 | 493 (491–496) | 10.0 (9.9–10.1) |
| bge-large | 335 | 1024 | 168 (168–169) | 18.3 (18.3–18.6) |
| Qwen3-Emb-0.6B | 596 | 1024 | 66 (66–66) | 52.2 (51.8–54.9) |

BM25 batch-1 query latency on CPU: 25.7 ms (23.7–27.7).

## 5.7. LLM judge validation

The LLM judge agreed poorly with the author's labels (Table 10). Cohen's $\kappa$ was 0.127 [−0.006, 0.261], with raw agreement 0.565. Weighted to the population, the judge labelled 0.886 of answers as supported, against 0.620 for the author, and its precision for the label supported was 0.636. Under the registered rule ($\kappa < 0.2$) we excluded the judge's faithfulness labels from all conclusions.

## 5.8. Summary of the predictions

Table 11 lists all six predictions. P3, P4 and P5 held. P1 failed in one of three new cells, P2 failed in both HotpotQA cells, and P6 failed. Where a verdict is not confirmatory, the table marks it.

**Table 8**
Approximate search (HNSW, $M$=32) vs exact search on the NQ sub-corpus, CPU.

| Encoder | Exact | | HNSW ef=64 | | | HNSW ef=128 | | |
|---|---|---|---|---|---|---|---|---|
| | Hit@5 | ms | Overlap | Hit@5 | ms | Overlap | Hit@5 | ms |
| MiniLM-L6 | 0.673 | 35.9 | 0.946 | 0.672 | 0.50 | 0.984 | 0.673 | 0.84 |
| bge-small | 0.729 | 36.3 | 0.904 | 0.726 | 0.49 | 0.964 | 0.727 | 0.89 |
| e5-base-v2 | 0.817 | 77.7 | 0.915 | 0.815 | 0.80 | 0.970 | 0.817 | 1.45 |

Overlap: mean share of the exact top 100 recovered. ms: median batch-1 query latency on 4 vCPUs.

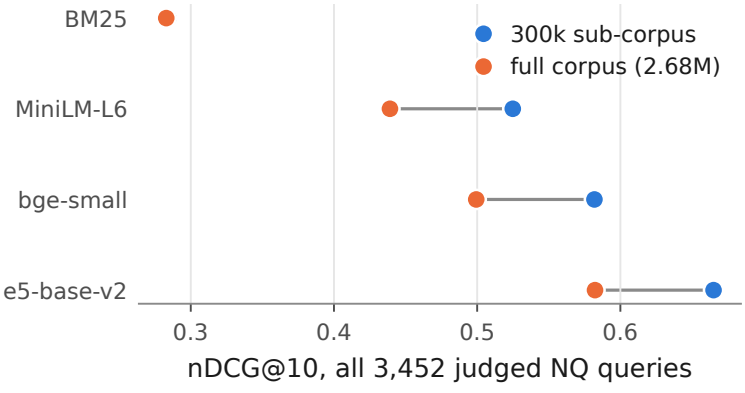



**Figure 6:** nDCG@10 on the 300,000-passage sub-corpus and on the full NQ corpus (all 3,452 judged queries).

**Table 9**
Sub-corpus (300k) vs full NQ corpus (2.68M), all 3,452 judged queries.

| Retriever | nDCG@10 sub | nDCG@10 full | Hit@5 sub | Hit@5 full |
|---|---|---|---|---|
| BM25 | 0.284 | 0.283 | 0.373 | 0.373 |
| MiniLM-L6 | 0.525 | 0.439 | 0.673 | 0.571 |
| bge-small | 0.582 | 0.499 | 0.729 | 0.630 |
| e5-base-v2 | 0.665 | 0.582 | 0.817 | 0.727 |

Kendall's $\tau$ between the two orderings = 1.00.

**Table 10**
Agreement between the author's labels and the LLM judge on 200 stratified items.

| | Judge SUPPORTED | Judge NOT_SUPPORTED |
|---|---|---|
| Human SUPPORTED | 65 | 50 |
| Human NOT_SUPPORTED | 37 | 48 |

Cohen's $\kappa$ = 0.127 [−0.006, 0.261]; raw agreement 0.565. Under the pre-registered rule ($\kappa < 0.2$) the judge results are excluded from the conclusions.

## 6. Discussion

### 6.1. What the transfer ratio shows

On single-hop NQ, a retrieval gain over a small encoder reached the answer at a rate of 0.16 to 0.27 EM points per Hit@5 point, with both readers. Three observations within our setup are consistent with this. One: the readers fell well short of their gold-passage bound even with good retrieval, so part of the loss lies in reading, not retrieval. Two: EM given a miss stayed far above closed-book EM. The NQ relevance labels are sparse, so many "misses" still contained an answer-bearing passage that was not labelled gold, or the

**Table 11**
Pre-registered predictions and their outcomes.

| ID | Scope | Result | Evidence |
|---|---|---|---|
| P1 | NQ, Qwen† | holds | 6 ratios; largest upper bound 0.455 (Qwen3-Emb-0.6B) |
| P1 | NQ, Llama | holds | 6 ratios; largest upper bound 0.427 (bge-small) |
| P1 | HotpotQA, Qwen | holds | 5 ratios; largest upper bound 0.996 (bge-large) |
| P1 | HotpotQA, Llama | **fails** | 5 ratios; largest upper bound 1.096 (bge-large) |
| P2 | NQ, Qwen† | holds | bge-large − e5: −0.006 ($p_{\text{Holm}}$ 1.000); Qwen3-Emb-0.6B − e5: −0.018 ($p_{\text{Holm}}$ 0.271) |
| P2 | NQ, Llama | holds | bge-large − e5: −0.004 ($p_{\text{Holm}}$ 1.000); Qwen3-Emb-0.6B − e5: −0.007 ($p_{\text{Holm}}$ 1.000) |
| P2 | HotpotQA, Qwen | **fails** | bge-large − e5: +0.028 ($p_{\text{Holm}}$ 0.005); Qwen3-Emb-0.6B − e5: −0.030 ($p_{\text{Holm}}$ 0.003) |
| P2 | HotpotQA, Llama | **fails** | bge-large − e5: +0.022 ($p_{\text{Holm}}$ 0.040); Qwen3-Emb-0.6B − e5: −0.042 ($p_{\text{Holm}}$ <0.001) |
| P3 | NQ, 4 retrievers | holds | Kendall's $\tau = 1.00$ |
| P4 | NQ, Qwen | holds | $k$=1 below $k$=5 for all 7 retrievers‡; ΔEM −0.126 to −0.078; all $p_{\text{Holm}}$ <0.001 |
| P5 | NQ, Qwen, 500 q. | holds | hard vs random negatives: ΔEM −0.074, $p$ <0.001 |
| P6 | NQ, Qwen, 500 q. | **fails** | gold at position 1/3/5: largest \|ΔEM\| 0.016; smallest $p_{\text{Holm}}$ 0.94 |

P1: every reportable EM transfer ratio has an upper 95% bound below 1. P2: no Holm-significant EM gain of bge-large or Qwen3-Emb-0.6B over e5-base-v2. P3: the full corpus keeps the nDCG@10 ordering. P4: EM at $k$=1 below $k$=5. P5: hard negatives lower EM. P6: gold position changes EM. †Not confirmatory: this cell was observed in Phase 1 of the study, before the predictions were registered. ‡Confirmatory for bge-small, bge-base, e5-base-v2 and Qwen3-Emb-0.6B only; the other three were observed before registration.

reader knew the answer. Hit@5 may therefore overstate the difference between encoders that matters for the reader. Three: EM given a hit fell slightly as encoders improved: the questions that a better encoder newly solves at the retrieval stage seem harder to answer. This last point is a hypothesis; we did not test which questions changed status.

On multi-hop HotpotQA, Hit@5 saturated while Both@5 did not. Most of the transfer came from higher EM given a hit, and the EM ranking of retrievers followed Both@5 exactly. A reasonable reading is that, for bridge questions, the second supporting passage carries the answer, and Hit@5 does not count it. Under this reading the large HotpotQA ratios are partly an artefact of the denominator: the true retrieval gain is larger than Hit@5 shows. The comparison questions, where both entities are named in the question, were close to their gold bound for every dense encoder. Practitioners who build multi-hop systems should measure retrieval with a metric that requires all needed evidence.

The two datasets therefore support different conclusions about the same encoders. A team that judges encoders by single-hop retrieval benchmarks will overrate the answer value of the strongest ones. A team with multi-hop questions may underrate it if it uses Hit@k.

## 6.2. Comparison with earlier work

Our NQ results agree with reports that retrieval gains do not carry over fully to answers (Salemi and Zamani, 2024; Mathur et al., 2026; Ghosh et al., 2026) and with the falling recall conversion rate reported by Peng et al. (2026). They do not contradict the strong correlations reported by Samuel et al. (2026): the EM ranking of retrievers followed their Hit@5 ranking closely in every cell (Spearman $\rho$ between 0.89 and 1.00). Correlation concerns the order of systems; the transfer ratio concerns the size of the gain. A metric can rank encoders correctly and still overstate how much answers improve.

The context experiment agrees with earlier findings on distraction. Top-ranked non-gold passages lowered EM compared with random passages (Cuconasu et al., 2024; Amiraz et al., 2025; Repantis et al., 2026), and random passages did not hurt. Gold position showed no detectable effect among five passages, in line with Cuconasu et al. (2025); with five short passages our setting is far from the long contexts in which position effects were originally shown (Liu et al., 2024). The near-identical results of exact and approximate search agree with Leto et al. (2024).

### 6.3. Practical implications

Within our setup, the choice of encoder mattered less for answers than for retrieval, while its cost differed by more than an order of magnitude. For single-hop questions, a base-size encoder such as e5-base-v2 or bge-base gave answers that no larger encoder improved detectably, while encoding 2.9 times faster than bge-large. For multi-hop questions, bge-large gave a small but detectable gain. Qwen3-Embedding-0.6B was the most expensive encoder and gave lower answers than e5-base-v2 on HotpotQA; its instruction and length settings may not suit this task, and we did not tune them. Retrieving ten passages instead of five raised EM for most retrievers and narrowed their differences, which is a cheaper lever than a larger encoder when the reader's context allows it. A cross-encoder helped the weak BM25 retrieval stage but not a strong dense one. Approximate search can replace exact search at no measurable cost in answers.

### 6.4. Alternative explanations

Several features of the design could produce low NQ transfer without a property of the encoders. EM undercounts correct answers (Kamalloo et al., 2023); the F1 and containment ratios were higher but showed the same contrast between datasets. The sub-corpus inflates dense retrieval scores, but the full-corpus check kept the order of four retrievers, and the inflation was similar for the three dense encoders tested (0.083 to 0.086 in nDCG@10). The readers were 4-bit quantised 7–8B models with one prompt; stronger readers may convert hits into answers at a different rate. The baseline encoder also matters: $T(R)$ against MiniLM-L6 measures gains above a fairly strong small encoder, and ratios against BM25 would differ.

## 7. Limitations

*Scope* The study covers two English Wikipedia question-answering datasets, six open encoders up to 0.6B parameters, two 4-bit 7–8B readers, one prompt and $k = 5$ for the transfer ratios. Results may differ for other domains, longer answers, larger or proprietary readers, other prompts or agentic pipelines. All timings come from one GPU type (Tesla T4) and one CPU configuration.

*Corpus and labels* Retrieval ran on 300,000-passage sub-corpora that contain every gold passage. The full-corpus check covers retrieval for four retrievers on NQ only; we did not rerun the readers on the full corpus or check HotpotQA. NQ relevance labels are sparse, so some "misses" contain the answer. The joined NQ set covers 2,255 of 3,452 judged queries and may differ in difficulty from the excluded queries.

*Metrics and statistics* EM is strict. We report F1 and answer containment, but did not judge correctness by hand. The transfer ratio is a ratio of means and becomes unstable when the retrieval gain is small; our reporting rule removes the worst cases, but the bge-small intervals on NQ still include zero. The decomposition and the Both@5 ranking are post hoc. Multiplicity is controlled within each family, not across all tests in the paper. Bootstrap intervals, F1, answer containment and the pooled model were checked by a rerun on a second computer, not by an independent implementation.

*Context experiment* The hard negatives are top-ranked BM25 passages without a gold label. Some may contain the answer; that would raise EM in the hard-negative condition and shrink the measured drop. We could not check this because the passage text is not stored with the outputs. The experiment used one reader and 500 questions.

*Judge and annotation* The LLM judge was validated against one annotator, the author. With $\kappa = 0.127$ the judge was excluded under the registered rule, so this limitation does not affect the conclusions, but no faithfulness result is reported.

*Two study phases* The predictions were registered in dated, version-controlled design documents in our own repository, not in a public registry. The Qwen–NQ cell and part of the $k$ ablation were run before the predictions were registered, and the software image changed between phases. Within-cell comparisons share one environment. For comparisons across phases, 396 of 397 identical prompts gave identical outputs.

## 8. Conclusion

We measured how much of a retrieval gain reaches the answer when a RAG system swaps one open embedding model for another. On single-hop NQ, between 0.16 and 0.27 EM points followed each Hit@5 point gained over MiniLM-L6, for both readers (RQ1). On multi-hop HotpotQA the ratio was larger, mainly because better encoders more often retrieved the second supporting passage, which Hit@5 does not count. Among dense encoders, bge-large gave a small EM gain over e5-base-v2 on HotpotQA and none detectable on NQ, and Qwen3-Embedding-0.6B gave lower EM (RQ2). Top-ranked non-gold passages lowered EM, gold position did not change it detectably, a reranker helped only the weak BM25 retrieval stage, and ten passages narrowed the gaps between retrievers (RQ3). Encoding throughput differed 27-fold, while approximate search and the sub-corpus did not change the order of retrievers (RQ4).

Within this setup, retrieval benchmarks overstate the answer value of stronger encoders on single-hop questions, and Hit@k understates it on multi-hop questions. We recommend that RAG studies report answer gains next to retrieval gains, with paired intervals, and use retrieval metrics that require all evidence a question needs. Future work should test larger readers, other domains and a full corpus for the readers, and should replace EM with validated human or model judgements.

## Data and code availability

All code, prompts, design documents with their registration dates, raw reader outputs, scored answers and analysis scripts are available for review at [[TBD: anonymised repository link, to be inserted by the author before submission]]. The datasets and models are public under their own licences; the exact revisions are listed in Appendix B. Every number in this paper is generated from the committed analysis output.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to write and test analysis and experiment code, to check references against bibliographic databases, and to draft and edit text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. Figures and tables were generated by scripts from the experimental data.

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**Table 12**
Pinned revisions (leading eight characters of the Hugging Face commit).

| Resource | Revision |
|---|---|
| sentence-transformers/all-MiniLM-L6-v2 | 1110a243 |
| BAAI/bge-small-en-v1.5 | 5c38ec7c |
| BAAI/bge-base-en-v1.5 | a5beb1e3 |
| BAAI/bge-large-en-v1.5 | d4aa6901 |
| intfloat/e5-base-v2 | f52bf8ec |
| Qwen/Qwen3-Embedding-0.6B | 97b0c614 |
| Qwen/Qwen2.5-7B-Instruct-AWQ | b2503754 |
| hugging-quants/Meta-Llama-3.1-8B-Instruct-AWQ-INT4 | db1f81ad |
| BAAI/bge-reranker-base | 2cfc18c9 |
| mistralai/Mistral-7B-Instruct-v0.3 (judge) | c170c708 |
| BeIR/nq | b7253e6c |
| BeIR/nq-qrels | 519acd4e |
| google-research-datasets/nq_open | 5dd9790a |
| BeIR/hotpotqa | a7e8bab2 |
| BeIR/hotpotqa-qrels | b15429e9 |
| hotpotqa/hotpot_qa | 1908d6af |

Zheng, L., Chiang, W.L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., et al., 2023. Judging LLM-as-a-Judge with MT-Bench and chatbot arena, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track. ArXiv:2306.05685.

## A. Reader prompts

Each prompt was rendered with the reader's own chat template. Passages appear in rank order as "[$i$] title" followed by the passage text on the next line, separated by blank lines.

*A1 (with passages)* System: *"You are a question answering assistant. Answer the question using the passages provided. If the passages do not contain the answer, give your best short answer anyway. Reply with the answer only, as a short phrase of a few words, with no explanation."* User: *"Passages: ... Question: {question} Answer:"*

*A2 (closed book)* System: *"You are a question answering assistant. Reply with the answer only, as a short phrase of a few words, with no explanation."* User: *"Question: {question} Answer:"*

The gold-passage condition uses A1 with the gold passages only.

## B. Models, data and software

Retrieval used bm25s 0.3.11 and faiss-cpu 1.15.1; readers used vLLM 0.30.0 with PyTorch 2.13.0 (CUDA 13.0) and Transformers 5.17.0, in half precision with a maximum model length of 4,096 tokens.

## C. Additional tables

Table 13 gives the bootstrap intervals for Table 1. Table 14 repeats the transfer ratios with token F1 and answer containment. Table 15 gives EM given a hit and given a miss, which enter Eq. 2.

**Table 13**
95% percentile bootstrap intervals (10,000 resamples over queries, seed 0) for the values in Table 1. Upper block: exact match per reader and dataset. Lower block: retrieval metrics, which do not depend on the reader.

| Condition | NQ Qwen | NQ Llama | HotpotQA Qwen | HotpotQA Llama |
|---|---|---|---|---|
| BM25 | [0.340, 0.380] | [0.321, 0.360] | [0.318, 0.359] | [0.315, 0.356] |
| MiniLM-L6 | [0.451, 0.492] | [0.427, 0.469] | [0.284, 0.325] | [0.278, 0.318] |
| bge-small | [0.463, 0.505] | [0.439, 0.481] | [0.372, 0.415] | [0.370, 0.413] |
| bge-base | [0.473, 0.513] | [0.456, 0.497] | [0.383, 0.426] | [0.394, 0.438] |
| bge-large | [0.482, 0.524] | [0.448, 0.489] | [0.402, 0.446] | [0.408, 0.453] |
| e5-base-v2 | [0.489, 0.530] | [0.452, 0.493] | [0.374, 0.417] | [0.387, 0.430] |
| Qwen3-Emb-0.6B | [0.471, 0.512] | [0.445, 0.487] | [0.345, 0.388] | [0.345, 0.388] |
| Closed book | [0.135, 0.164] | [0.180, 0.213] | [0.143, 0.174] | [0.133, 0.165] |
| Gold passages | [0.633, 0.673] | [0.593, 0.633] | [0.607, 0.649] | [0.602, 0.645] |
| Retrieval | NQ Hit@5 | | HotpotQA Hit@5 | HotpotQA Both@5 |
| BM25 | [0.343, 0.383] | | [0.816, 0.849] | [0.244, 0.283] |
| MiniLM-L6 | [0.645, 0.683] | | [0.799, 0.834] | [0.212, 0.248] |
| bge-small | [0.710, 0.747] | | [0.944, 0.963] | [0.519, 0.563] |
| bge-base | [0.753, 0.788] | | [0.949, 0.967] | [0.550, 0.594] |
| bge-large | [0.779, 0.812] | | [0.952, 0.969] | [0.599, 0.642] |
| e5-base-v2 | [0.800, 0.832] | | [0.941, 0.960] | [0.522, 0.567] |
| Qwen3-Emb-0.6B | [0.726, 0.762] | | [0.934, 0.955] | [0.422, 0.467] |

**Table 14**
Transfer ratios with token F1 and answer containment in place of EM (same queries, baseline and reporting rule as Table 2).

| Retriever | NQ Qwen | NQ Llama | HotpotQA Qwen | HotpotQA Llama |
|---|---|---|---|---|
| *Token F1* | | | | |
| BM25 | 0.43 [0.37, 0.48] | 0.43 [0.37, 0.49] | n.e. | n.e. |
| bge-small | 0.26 [0.02, 0.49] | 0.23 [0.00, 0.45] | 0.76 [0.62, 0.92] | 0.85 [0.70, 1.01] |
| bge-base | 0.26 [0.11, 0.40] | 0.33 [0.19, 0.47] | 0.79 [0.65, 0.95] | 0.95 [0.80, 1.11] |
| bge-large | 0.27 [0.15, 0.39] | 0.25 [0.13, 0.36] | 0.97 [0.82, 1.14] | 1.10 [0.95, 1.28] |
| e5-base-v2 | 0.30 [0.20, 0.39] | 0.26 [0.17, 0.36] | 0.77 [0.62, 0.92] | 0.97 [0.82, 1.13] |
| Qwen3-Emb-0.6B | 0.26 [0.07, 0.45] | 0.29 [0.11, 0.47] | 0.52 [0.38, 0.66] | 0.66 [0.52, 0.81] |
| *Answer containment* | | | | |
| BM25 | 0.45 [0.39, 0.52] | 0.46 [0.39, 0.52] | n.e. | n.e. |
| bge-small | 0.27 [0.01, 0.52] | 0.32 [0.07, 0.57] | 0.68 [0.53, 0.84] | 0.75 [0.61, 0.92] |
| bge-base | 0.34 [0.18, 0.49] | 0.38 [0.23, 0.53] | 0.72 [0.57, 0.88] | 0.90 [0.75, 1.07] |
| bge-large | 0.34 [0.21, 0.46] | 0.33 [0.21, 0.46] | 0.88 [0.73, 1.04] | 1.01 [0.86, 1.18] |
| e5-base-v2 | 0.36 [0.26, 0.47] | 0.33 [0.23, 0.43] | 0.71 [0.57, 0.87] | 0.91 [0.75, 1.07] |
| Qwen3-Emb-0.6B | 0.28 [0.08, 0.48] | 0.39 [0.20, 0.60] | 0.49 [0.35, 0.64] | 0.58 [0.45, 0.74] |

**Table 15**
Exact match given a hit and given a miss at Hit@5, $P(\mathrm{EM} \mid \mathrm{hit})$ / $P(\mathrm{EM} \mid \mathrm{miss})$, on the generation queries.

| Retriever | NQ Qwen | NQ Llama | HotpotQA Qwen | HotpotQA Llama |
|---|---|---|---|---|
| BM25 | 0.632 / 0.204 | 0.591 / 0.197 | 0.381 / 0.129 | 0.382 / 0.105 |
| MiniLM-L6 | 0.584 / 0.249 | 0.554 / 0.237 | 0.350 / 0.098 | 0.348 / 0.071 |
| bge-small | 0.568 / 0.259 | 0.542 / 0.239 | 0.411 / 0.043 | 0.409 / 0.043 |
| bge-base | 0.564 / 0.257 | 0.538 / 0.270 | 0.419 / 0.060 | 0.432 / 0.048 |
| bge-large | 0.568 / 0.252 | 0.533 / 0.220 | 0.438 / 0.076 | 0.446 / 0.063 |
| e5-base-v2 | 0.569 / 0.244 | 0.525 / 0.237 | 0.415 / 0.030 | 0.428 / 0.040 |
| Qwen3-Emb-0.6B | 0.572 / 0.260 | 0.538 / 0.256 | 0.383 / 0.090 | 0.383 / 0.072 |

Closed-book EM for comparison: NQ 0.149 (Qwen), 0.197 (Llama); HotpotQA 0.158 (Qwen), 0.148 (Llama).